



Data science in auditing

Master Accountancy

Case “DigiJazz”

1

DigiJazz

During this course, we make use of the **DigiJazz GL Dataset**, which contains all General Ledger (GL) Transactions for DigiJazz for one year, obtained from their XML Auditfile Financieel (XAF).

Introduction DigiJazz

DigiJazz.nl is an internationally recognized e-commerce company specializing in the online sale of jazz music. It was founded in 2015 by Dean Mus and Rob Tone. The company provides a 24/7 platform for customers worldwide to download jazz music.

The start-up has been extremely successful and has experienced rapid growth ever since. In 2017, DigiJazz also began selling vinyl records. This was a great success, as many jazz enthusiasts appreciate the nostalgia of vinyl. Consequently, revenue has significantly increased annually since 2018.



Online sales in the webshop

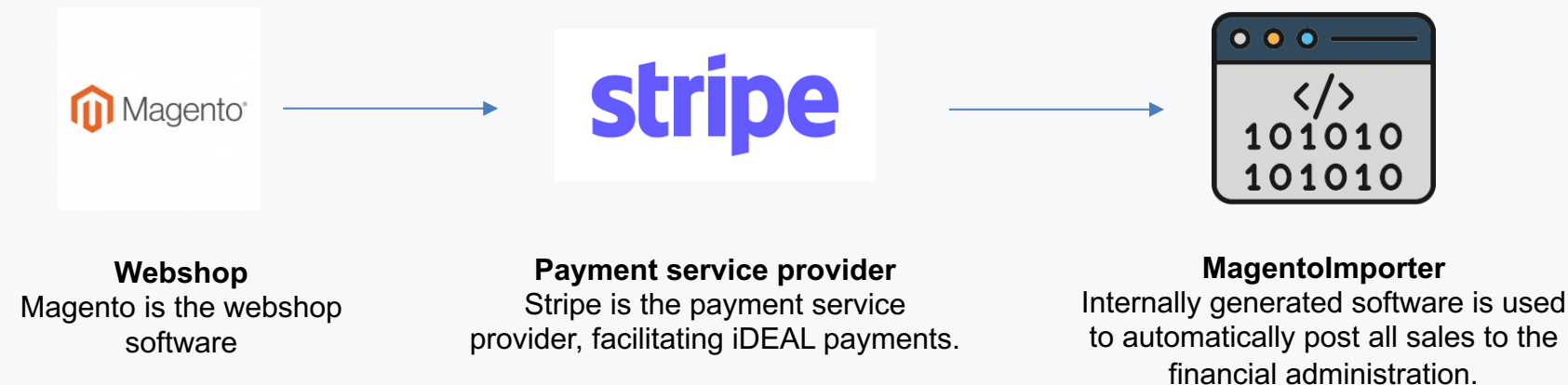
Sales of online music and vinyl records occur exclusively through the website DigiJazz.nl. Within this web environment, customers can search for, listen to, order, and download their favorite music. Transactions are also recorded and processed in this system.

DigiJazz has the following revenue categories:

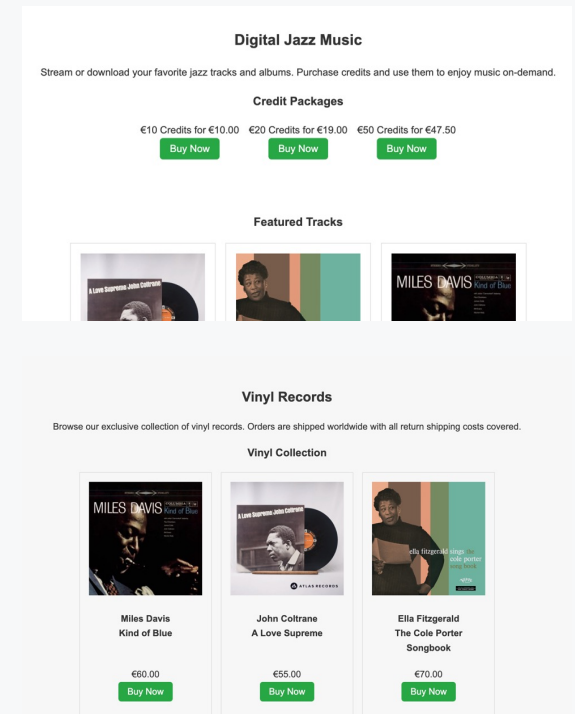
1. Online streaming and downloading of digital jazz music;
2. Sales and delivery of jazz vinyl records.

You can see screenshots from the DigiJazz website on the right.

Processing of the orders: Order processing is (financially) fully automated, as displayed below:



Screenshots from the Digijazz website:



Sales prices and margins

An essential principle for performing risk assessment activities is that the auditor *gains an understanding of the entity and its environment* (see paragraph 3.3 of NBA Handreiking 1141).

We started the engagement with a valuable conversation with management. They explained us the key aspects of the business. We can use this understanding in our data analytics.

Management has indicated:

- Typical digital music sales transactions **have a transaction value of less than EUR 30**
- The majority of vinyl sales transactions have a transaction value **between EUR 45 and EUR 75**.
- The cost of sales consists of purchased vinyl (inventory) or royalties payable on sales. **The margin is the same on all orders: 35%**. There should be no sales with a lower margin.

This aligns with the knowledge the auditor has based on audits from previous years.

Digital Jazz Music

Stream or download your favorite jazz tracks and albums. Purchase credits and use them to enjoy music on-demand.

Credit Packages

€10 Credits for €10.00	€20 Credits for €19.00	€50 Credits for €47.50
Buy Now	Buy Now	Buy Now

Featured Tracks



Vinyl Records

Browse our exclusive collection of vinyl records. Orders are shipped worldwide with all return shipping costs covered.

Vinyl Collection



Miles Davis
Kind of Blue

€60.00
[Buy Now](#)



John Coltrane
A Love Supreme

€55.00
[Buy Now](#)



Ella Fitzgerald
The Cole Porter
Songbook

€70.00
[Buy Now](#)

Dataset introduction

The dataset contains all general ledger transactions for DigiJazz B.V. for the current year.

GL Data

Since we are looking at an auditfile, our dataset consists of debit and credit lines. You can recognize each transaction based on the 'Booknr' column.

Example:

```
[11] gl_data[gl_data['Booknr'] == 'SJ079206'][['Description', 'Booknr', 'Account', 'Debit/Credit', 'Amount']]
```

	Description	Booknr	Account	Debit/Credit	Amount
0	Digijazz.nl Order - Mp3603	SJ079206	8000 Revenue digital music	Credit	9.00
87197	Digijazz.nl Order - Mp3603	SJ079206	1300 Debtors	Debit	10.89
565582	Digijazz.nl Order - Mp3603	SJ079206	1500 VAT Payable	Credit	1.89

Mapping

The GL accounts and mapping from last year are presented on the right. In the current year, limited changes have been made.

Balance		
AccountType	Account	Amount PY
Asset	1101 Bank account *313	6,260,418
Asset	1200 Inventory Vinyl	1,808,900
Asset	1300 Debtors	8,416,356
Asset	1520 VAT Recoverable	251,303
Liability	1500 VAT Payable	730,263
Liability	1600 Creditors	3,295,580
Liability	1670 Net wages	562,762
Liability	1780 Payable reservation for vacation allowance	28,800
Liability	1910 Accruals Payable	190,720
Liability	1950 Payroll taxes payable	216,000
Liability	1980 Invoices to be received digital music	458,170

P&L		
AccountType	Account	Amount PY
Revenue	8000 Revenue digital music	702,916
Revenue	8001 Revenue Vinyl	2,774,979
COGS	7000 Cost of sales digital music	458,170
COGS	7001 Cost of sales Vinyl	1,808,900
Expense	4000 Gross wages	376,800
Expense	4040 Compensation commuting	12,581
Expense	4080 Vacation allowance costs	28,800
Expense	5510 Purchases	1,196,681
Expense	5511 Rent Expense	14,400
Expense	5512 Accounting and Audit Fees	80,000
Expense	5513 Utilities	7,200
Expense	5514 Car Lease	62,400
Expense	5520 IT Hosting	20,757

Data dictionary

The data contains the following columns:

- **Date:** The date of the transaction.
- **Period:** The accounting period in YYYY-MM format, e.g. 2024-01.
- **Description:** A brief explanation or details about the transaction.
- **JournalCode:** Code of the journal (e.g. SJ for the Sales Journal).
- **JournalDesc:** Description of the journal.
- **JournalType:** Type/category of the journal (e.g., Sales, Cash, Purchases).
- **Booknr:** Reference number for the bookkeeping entry, unique for each transaction.
- **AccountCode:** Code representing the General Ledger (GL) Account involved in the transaction.
- **AccountDesc:** Title of the GL Account.
- **AccountType:** Type/category of the GL Account (e.g., Revenue, Asset).
- **Debit/Credit:** Indicates if the amount is a debit or credit.
- **Amount:** The monetary value of the transaction (in EUR)
- **Product ID:** Identifier for the product involved (if applicable: only for sales transactions).
- **User:** The system or individual responsible for creating the transaction entry.
- **Relation_id:** Identifier for the related customer or party (for example the debtor/creditor).
- **Relation_name:** Name of the related customer or party (for example the debtor/creditor).
- **Source_id:** Source system reference or identifier.
- **Account:** Full GL account name combining the code and description.